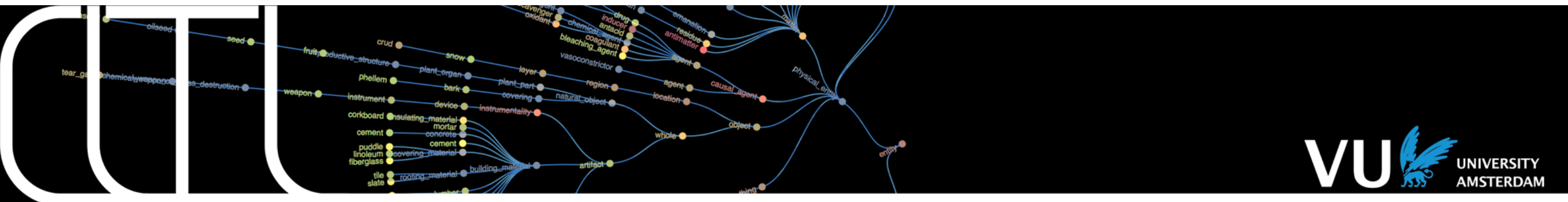


# Text Mining 2026



## Lecture 6: Topic Modelling and Text Classification Ilia Markov



# What is a topic?

---

- Topic = main area of interest that a text is about (not a strict definition):
  - text can be about **events** (9/11, invasion in Ukraine: many different spreading events, difficult to figure out what belongs to the topic + insignificant/important events), **activities** -> socially defined events (soccer, elections, Olympics, traffic, tourism)), **situations** (pandemic), **people**, **organisations**, **places**, **abstract concepts** (freedom of speech, democracy, consumer trust, corruption) -> each of them can be a topic of a text (open-ended)
- There is no a priori definition of all topics:
  - it is subjective, cultural and topics as the world change continuously
  - information brokers (LexisNexis, Bloomberg) make it their business to create a personal/company profile of interest and deliver relevant news on demand
- but some people try anyway with a reusable definition.... so let us look at news

# International Press Telecommunications Counsel

- Works on standardized taxonomy of subjects (topics)
- Media Topics subject taxonomy: 1,200 terms, 17 top levels, 4 levels deep to describe all news articles in terms of topics they cover
- Available in multiple languages
- <https://iptc.org> for information on each topic
- MediaTopic ID: first digits to indicate the main category of a topic; further positions for subdivisions of topics (4 levels deep), e.g., ‘conflict, war and peace’, etc. But never complete!

## IPTC Media Topic NewsCodes as of 2021-02-18 (language: en-GB)

Expand All Collapse All View language versions: [Arabic](#) | [Chinese \(分类\)](#) | [Danish](#) | [English \(en-GB\)](#) | [English \(en-US\)](#) | [French](#) | [German](#) | [Norwegian](#) | [Portuguese](#) | [Portuguese \(Brazil\)](#) | [Spanish](#) | [Swedish](#)

| MediaTopic ID                     | Name (en-GB)                              | Definition (en-GB)                                                                                                                                                                                                        |
|-----------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ▶ <a href="#">medtop:01000000</a> | arts, culture, entertainment and media    | All forms of arts, entertainment, cultural heritage and media                                                                                                                                                             |
| ▶ <a href="#">medtop:16000000</a> | conflict, war and peace                   | Acts of socially or politically motivated protest or violence, military activities, geopolitical conflicts, as well as resolution efforts                                                                                 |
| ▶ <a href="#">medtop:02000000</a> | crime, law and justice                    | The establishment and/or statement of the rules of behaviour in society, the enforcement of these rules, breaches of the rules, the punishment of offenders and the organisations and bodies involved in these activities |
| ▶ <a href="#">medtop:03000000</a> | disaster, accident and emergency incident | Man made or natural event resulting in loss of life or injury to living creatures and/or damage to inanimate objects or property                                                                                          |
| ▶ <a href="#">medtop:04000000</a> | economy, business and finance             | All matters concerning the planning, production and exchange of wealth.                                                                                                                                                   |
| ▶ <a href="#">medtop:05000000</a> | education                                 | All aspects of furthering knowledge of human individuals from birth to death.                                                                                                                                             |
| ▶ <a href="#">medtop:06000000</a> | environment                               | All aspects of protection, damage, and condition of the ecosystem of the planet earth and its surroundings.                                                                                                               |
| ▶ <a href="#">medtop:07000000</a> | health                                    | All aspects pertaining to the physical and mental welfare of living beings                                                                                                                                                |
| ▶ <a href="#">medtop:08000000</a> | human interest                            | Items about individuals, groups, animals, plants or other objects with a focus on emotional facets                                                                                                                        |
| ▶ <a href="#">medtop:09000000</a> | labour                                    | Social aspects, organisations, rules and conditions affecting the employment of human effort for the generation of wealth or provision of services and the economic support of the unemployed.                            |
| ▶ <a href="#">medtop:10000000</a> | lifestyle and leisure                     | Activities undertaken for pleasure, relaxation or recreation outside paid employment, including eating and travel.                                                                                                        |
| ▶ <a href="#">medtop:11000000</a> | politics                                  | Local, regional, national and international exercise of power, or struggle for power, and the relationships between governing bodies and states.                                                                          |
| ▶ <a href="#">medtop:12000000</a> | religion and belief                       | All aspects of human existence involving theology, philosophy, ethics and spirituality.                                                                                                                                   |
| ▶ <a href="#">medtop:13000000</a> | science and technology                    | All aspects pertaining to human understanding of, as well as methodical study and research of natural, formal and social sciences, such as astronomy, linguistics or economics                                            |
| ▶ <a href="#">medtop:14000000</a> | society                                   | The concerns, issues, affairs and institutions relevant to human social interactions, problems and welfare, such as poverty, human rights and family planning                                                             |
| ▶ <a href="#">medtop:15000000</a> | sport                                     | Competitive activity or skill that involves physical and/or mental effort and organisations and bodies involved in these activities                                                                                       |
| ▶ <a href="#">medtop:17000000</a> | weather                                   | The study, prediction and reporting of meteorological phenomena                                                                                                                                                           |

# International Press Telecommunications Counsel

- Expand ‘crime, law and justice’ to subcategories, 02 subcategory is divided into various sublayers (‘animal abuse’, ‘sex crime’, etc.)
- New topics (‘cybercrime’).
- Topics develop, this is a continuous process

## IPTC Media Topic NewsCodes as of 2021-02-18 (language: en-GB)

Expand All Collapse All View language versions: [Arabic](#) | [Chinese \(分类\)](#) | [Danish](#) | [English \(en-GB\)](#) | [English \(en-US\)](#) | [French](#) | [German](#) | [Norwegian](#) | [Portuguese](#) | [Portuguese \(Brazil\)](#) | [Spanish](#) | [Swedish](#)

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| ▼ <a href="#">medtop:20000082</a> | crime                                  | Violations of laws by individuals, companies or organisations                                                                                                                                                             |
| <a href="#">medtop:20001314</a>   | animal abuse                           | The infliction of cruelty or neglect onto animals by humans                                                                                                                                                               |
| <a href="#">medtop:20000083</a>   | arson                                  | The intentional setting of fires with criminal intent                                                                                                                                                                     |
| ▼ <a href="#">medtop:20000084</a> | assault                                | Physical crime against a person, including battery, brawls and threat of bodily harm                                                                                                                                      |
| <a href="#">medtop:20000085</a>   | sex crime                              | Sexual crimes against a person, such as rape, harassment, molestation or groping                                                                                                                                          |
| ▼ <a href="#">medtop:20000087</a> | corporate crime                        | Crimes committed by a corporation or by individuals acting on behalf of a corporation                                                                                                                                     |
| <a href="#">medtop:20001194</a>   | accounting crime                       | The intentional misrepresentation or alteration of accounting records regarding sales, revenues, expenses and other factors                                                                                               |
| <a href="#">medtop:20000088</a>   | anti-trust crime                       | Violations of laws aimed at corporations to promote fair competition for the benefit of consumers                                                                                                                         |
| <a href="#">medtop:20000089</a>   | breach of contract                     | The breaking of terms set in a contractual agreement                                                                                                                                                                      |
| <a href="#">medtop:20000090</a>   | embezzlement                           | The intentional theft or misuse of money left in one's care or belonging to one's employer                                                                                                                                |
| <a href="#">medtop:20000091</a>   | insider trading                        | Trading of stocks by persons having access to internal information not made public at the time                                                                                                                            |
| <a href="#">medtop:20000092</a>   | restraint of trade                     | Interference in free competition in business and trade                                                                                                                                                                    |
| ▼ <a href="#">medtop:20000093</a> | corruption                             | Dishonest actions by a person in power in return for pecuniary or personal gain                                                                                                                                           |
| <a href="#">medtop:20000094</a>   | bribery                                | The giving of money or favours in order to influence the judgment or conduct of a person in a position of power                                                                                                           |
| <a href="#">medtop:20000086</a>   | cyber crime                            | Criminal activities carried out by means of computers or the Internet, such as hacking, phishing scams and identity theft                                                                                                 |
| ▼ <a href="#">medtop:20000095</a> | drug related crimes                    | Illegal activities involving illicit substances, including buying, distribution, selling and smuggling                                                                                                                    |
| <a href="#">medtop:20000096</a>   | drug trafficking                       | Dealing in illicit often harmful substances                                                                                                                                                                               |
| ▼ <a href="#">medtop:20000097</a> | fraud                                  | Intentional deception for personal or financial gain                                                                                                                                                                      |
| <a href="#">medtop:20001195</a>   | tax evasion                            | The illegal evasion of taxes by individuals, corporations, and trusts                                                                                                                                                     |
| <a href="#">medtop:20000098</a>   | hijacking                              | Taking over a transportation vehicle by force                                                                                                                                                                             |
| <a href="#">medtop:20000099</a>   | homicide                               | The killing of one person by another, including murder and manslaughter                                                                                                                                                   |
| <a href="#">medtop:20001283</a>   | human smuggling and trafficking        | The illegal transporting of people from one country or area to another. When the result is forced labour or sexual exploitation, the crime is considered trafficking.                                                     |
| <a href="#">medtop:20000100</a>   | kidnapping                             | To seize and detain a person against that person's will by unlawful threat, force or fraud                                                                                                                                |

# What is (not) a topic?

---

- **Genres are not topics but conventional styles of communication:**
  - burst of tweets cluttered in time
  - news articles, blogs reporting on yesterday's events
  - educational/scientific texts, Wikipedia
  - conversations, debates (interaction)
  - product reviews
  - books, songs, movies
  - historical analysis
  - tweets, fora, Reddit, Whatsapp, Facebook, Instagram, TikTok
- **Short, long, noisy, time-dependent, dynamic, large volumes, etc.**
- **All of the above impact the technology used to identify topics**

# Text and topics

---

- Tweet: short, referential:

The new US president has said he believes that vaccines are harmful and has repeatedly and erroneously suggested that they cause autism

- Topics: vaccines, autism, Trump, radicalisation? - difficult to figure out w/o wider context
- News: long, stories with several events, more self-contained but still referential to the world
- Referential implies related to a context (here and now) but could be to a fictional world, e.g. GoT, StarTrek



# Text and topics

---

- **Tweet:** short, referential relying on context, **News:** longer stories, referential but more self-contained
- Reuters: <http://www.daviddlewis.com/resources/testcollections/reuters21578/>
  - LDC FOOD AID NEEDS DECLINE IN 1986/87 - USDA
  - Total [food aid](#) needs in 69 of the least developed countries declined in 1986/87, as requirements fell in many countries in [Africa](#), the [Middle East](#) and [Asia](#), the U.S. Agriculture Department said.
  - In a summary of its World Agriculture Report, the department said [grain production](#) in sub-Saharan Africa was a record high in 1986, with gains in almost every country. However, [food needs](#) in Central America rose, worsened by [drought-reduced crops](#) and [civil strife](#). Record [wheat](#) production in 1986/87 is pushing global [wheat](#) consumption for [food](#) to a new high, and higher yielding varieties have been particularly effective where spring [wheat](#) is a common [crop](#), it said.
  - However, many developing countries in tropical climates, such as [Sub-Saharan Africa](#), [Southeast Asia](#), and [Central America](#), are not well adapted for [wheat](#) production, and improved varieties are not the answer to rising [food needs](#), the department said.
  - World per capita consumption of [vegetable oil](#) will rise in 1986/87 for the third straight year. [Soybean oil](#) constitutes almost 30 pct of [vegetable oil](#) consumption, while [palm oil](#) is the most traded, the department said.
- **Words highlighted in red relate to the topic**
- **Topics:** grain, wheat, oilseed, soybean, veg-oil = **labels for a supervised classification**
- **Question:** are there other topics that could have been mentioned?

# Text and topics

---

- **Tweet:** short, referential relying on context, **News:** longer stories, referential but more self-contained
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  - In a summary of its World Agriculture Report, the department said grain production in sub-Saharan Africa was a record high in 1986, with gains in almost every country. However, food needs in Central America rose, worsened by drought-reduced crops and civil strife. Record wheat production in 1986/87 is pushing global wheat consumption for food to a new high, and higher yielding varieties have been particularly effective where spring wheat is a common crop, it said.
  - However, many developing countries in tropical climates, such as Sub-Saharan Africa, Southeast Asia, and Central America, are not well adapted for wheat production, and improved varieties are not the answer to rising food needs, the department said.
  - World per capita consumption of vegetable oil will rise in 1986/87 for the third straight year. Soybean oil constitutes almost 30 pct of vegetable oil consumption, while palm oil is the most traded, the department said.
- **Words in red relate to the topic**
- **Topics:** grain, wheat, oilseed, soybean, veg-oil
- **Other potential topics that are missing:** food, food aid, food needs, drought, crops, civil strife, palm oil, food consumption
- **What is the level of granularity to assign the topics? Depends on data usage.**



# Reuters-21578, 120 categories

|                  |              |               |                    |               |                  |
|------------------|--------------|---------------|--------------------|---------------|------------------|
| 1 bfr            | 2 linseed    | 4 pork-belly  | 17 lumber          | 43 palm-oil   | 120 soybean      |
| 1 castorseed     | 2 rye        | 4 tapioca     | 17 sunseed         | 44 zinc       | 130 nat-gas      |
| 1 citruspulp     | 2 sun-meal   | 6 coconut     | 18 income          | 51 meal-feed  | 135 gold         |
| 1 corn-oil       | 2 wool       | 6 potato      | 21 housing         | 51 rubber     | 137 veg-oil      |
| 1 cottonseed     | 3 can        | 6 propane     | 21 stg             | 54 barley     | 145 coffee       |
| 1 cruzado        | 3 copra-cake | 7 coconut-oil | 25 heat            | 63 alum       | 163 gnp          |
| 1 dkr            | 3 cotton-oil | 7 instal-debt | 25 soy-oil         | 63 cotton     | 184 sugar        |
| 1 hk             | 3 dfl        | 7 inventories | 27 hog             | 65 gas        | 190 money-supply |
| 1 lin-meal       | 3 f-cattle   | 7 naphtha     | 27 retail          | 65 ipi        | 192 oilseed      |
| 1 peseta         | 3 lit        | 8 jet         | 27 soy-meal        | 67 iron-steel | 217 dlr          |
| 1 rape-meal      | 3 nkr        | 8 rape-oil    | 28 fuel            | 67 rice       | 254 corn         |
| 1 red-bean       | 3 palladium  | 8 sun-oil     | 29 orange          | 69 yen        | 305 ship         |
| 1 ringgit        | 3 palmkernel | 9 l-cattle    | 32 strategic-metal | 75 carcass    | 306 wheat        |
| 1 rupiah         | 3 rand       | 10 groundnut  | 32 wpi             | 76 cocoa      | 513 interest     |
| 1 skr            | 3 saudriyal  | 11 nickel     | 33 tin             | 76 jobs       | 552 trade        |
| 2 castor-oil     | 3 sfr        | 12 platinum   | 35 lead            | 78 copper     | 628 grain        |
| 2 cornglutenfeed | 4 austdlr    | 14 oat        | 35 rapeseed        | 84 reserves   | 634 crude        |
| 2 fishmeal       | 4 cpu        | 15 dmK        | 35 sorghum         | 112 cpi       | 801 money-fx     |
| 2 groundnut-oil  | 4 nzdlr      | 15 tea        | 37 silver          | 114 livestock | 2448 acq         |
| 2 lin-oil        | 4 plywood    | 17 lei        | 41 pet-chem        | 116 bop       | 3987 earn        |

What does the choice of topics tell us about the people that provided this data?

# How to assign a topic to a text?

---

- Topic classification can be seen as a form of text classification (broad term to cover any association between a label and a text, e.g., sentiment, genre, language)
- Given labels assigned to the document level:
  - Supervised text classification
    - Explicit feature extraction: BoW, word n-grams, character n-grams, etc.
    - Fine-tuning of language models
  - Unsupervised clustering: use an inner representation of texts and group texts based on the similarity of
    - Words/Tokens
    - Word Embeddings
- Units: complete books, documents, short texts, sentences (e.g., titles); depends on the final use
- Multi-label classification: one-against-all
- How to represent and communicate topics:
  - Keyword extraction from document clusters (typically tf-idf is used)

# Literature

---

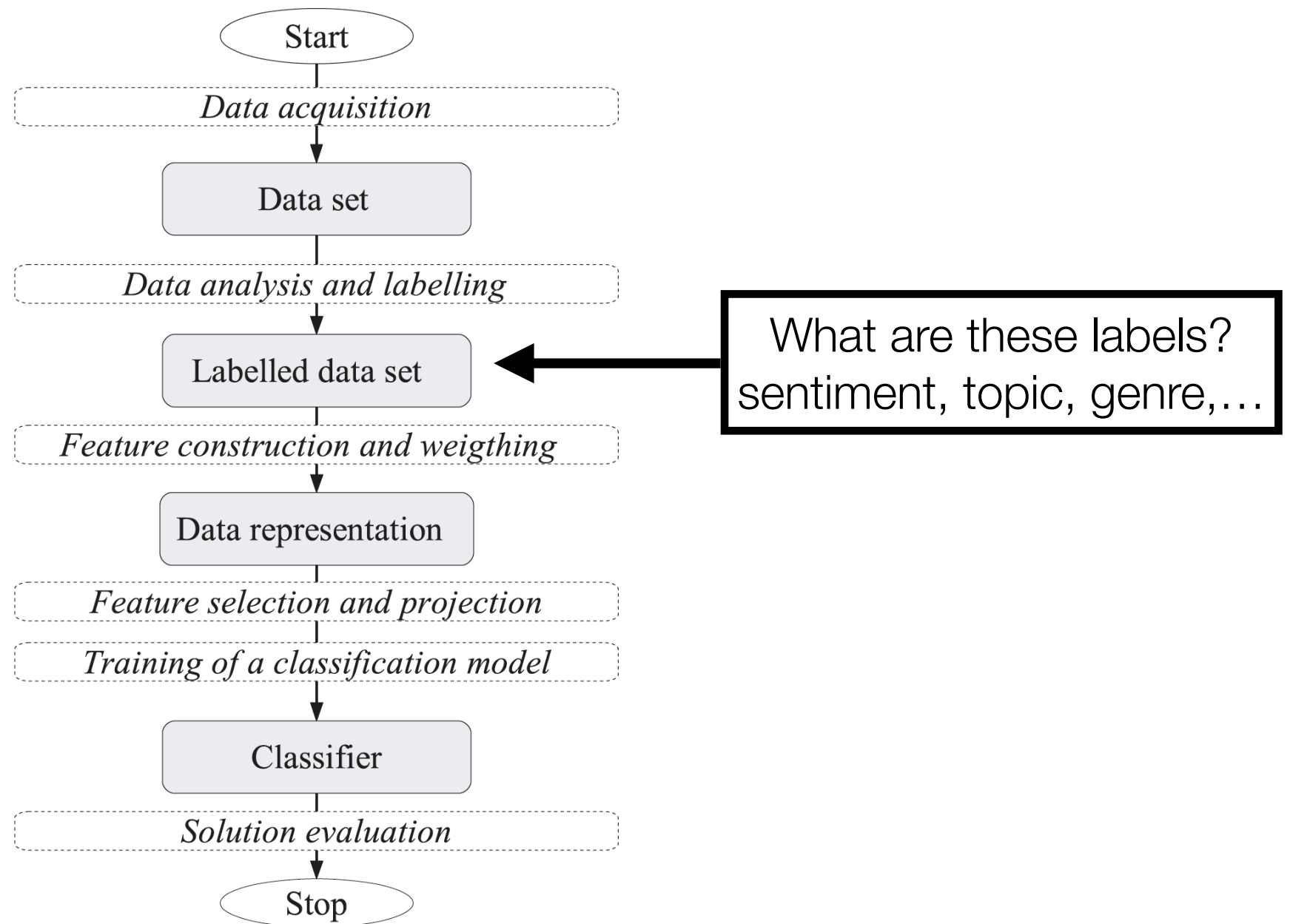
- Saigal, P., Khanna, V. Multi-category news classification using Support Vector Machine based classifiers. *SN Appl. Sci.* 2, 458 (2020). (Skip mathematical explanations of SVM models, focus on experiment setup and evaluation)
- Churchill, Rob, and Lisa Singh. "The Evolution of Topic Modeling." *ACM Computing Surveys (CSUR)* (2021) (Survey paper on development of topic modeling: high-level introduction w/ challenges, skip section 5 on variants of a basic algorithm)
- I. Vayansky and S.A.P. Kumar, A review of topic modeling methods. *Information Systems* 94 (2020) 101582 (Mathematical background of LDA variants)
- Sun, Chi, Xipeng Qiu, Yige Xu, and Xuanjing Huang. "How to fine-tune BERT for text classification?." In *China National Conference on Chinese Computational Linguistics*, pp. 194-206. Springer, Cham, 2019

# Supervised topic classification

---

- Training data with texts and topic labels
- Bag-of-words representation of a text
- TD\*IDF weights as the information value of a word for a document
- Dimensionality Reduction (Latent Semantic Analysis): grouping words with similar distributions to decrease data sparseness
  - e.g. king, queen, rook, pawn, bishop and knight vote for chess so group them together so that presence of any of these words counts for a concept

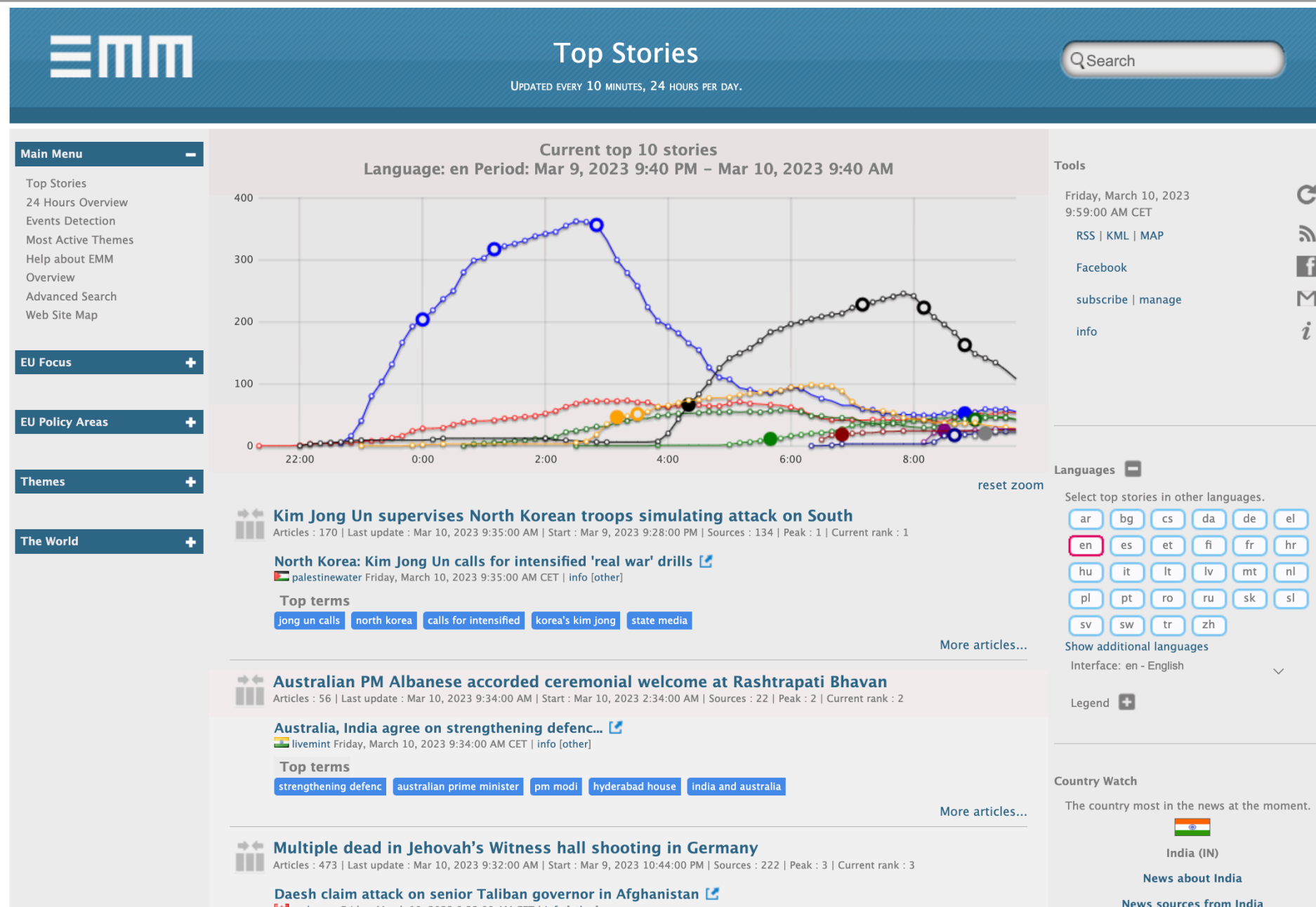
# Supervised topic classification



**Fig. 1.** Flowchart of the text classification process with the state-of-the-art elements.

Mirończuk, Marcin Michał, and Jarosław Protasiewicz, *A recent overview of the state-of-the-art elements of text classification*. *Expert Systems with Applications* 106 (2018): 36-54.

# Newsbrief: European Joint Research Centre



- <https://emm.newsbrief.eu/NewsBrief/clusteredition/en/latest.html>  
Daily information about what are the actual topics in the discussions that are currently in the news.



# Supervised topic classification

---

- Steinberger, Ebrahim & Turchi, JRC EuroVoc Indexer JEX – A freely available multi-label categorisation tool
- <https://eur-lex.europa.eu/browse/eurovoc.html> (labels from the European thesaurus of topics for European politicians)
- JRC-Acquis, a manually labeled legislative texts in 22 official EU languages:
  - <https://opus.nlpl.eu/JRC-Acquis.php>
  - <https://wt-public.emm4u.eu/Acquis/JRC-Acquis.3.0/corpus/>

# Supervised topic classification

---

- Advantages:
  - Control over the topics
  - High performance (e.g., sufficient training data for relevant topics)
- Disadvantages:
  - New topics cannot be detected
  - Biased towards distribution of the training data
  - Manual effort to maintain the topics and training data

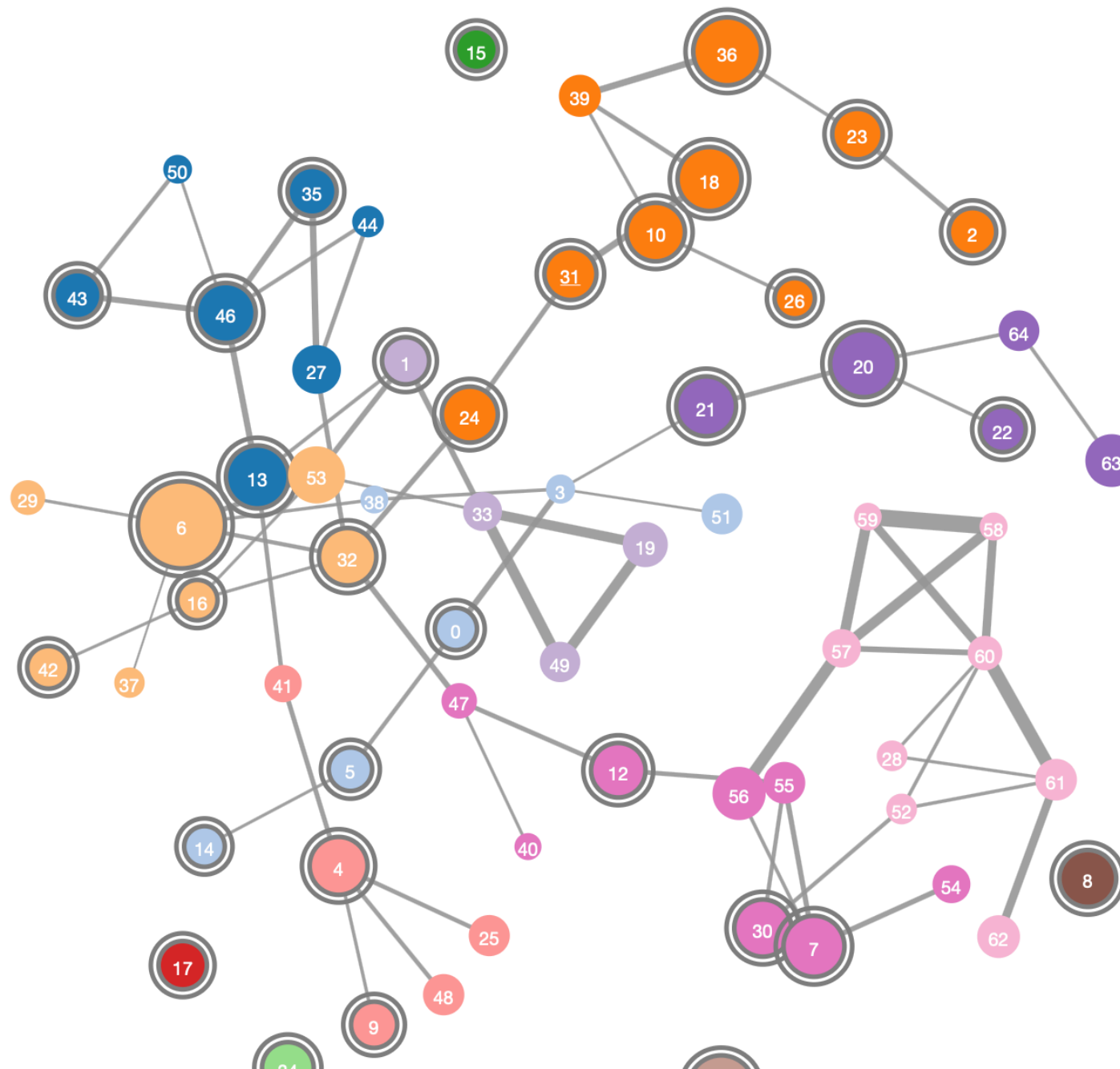
# Unsupervised topic modeling

---

- Topic modeling is a clustering task that groups documents on the basis of their shared word associations.
- Why? In case there is no labeled data or labels are not appropriate
- Assumes that any set of documents can be split into groups or clusters that use similar words.
- How to label clusters?
  - Keywords can be extracted from these clusters that are most distinctive (TD\*IDF)

# National Science Agenda

- **11K** questions with titles, descriptions and keywords
- Term extraction & hierarchical clustering
- <http://i.amcat.nl/renwa/index.html>



## Selected cluster

### Selected cluster:31

#Questions: 133

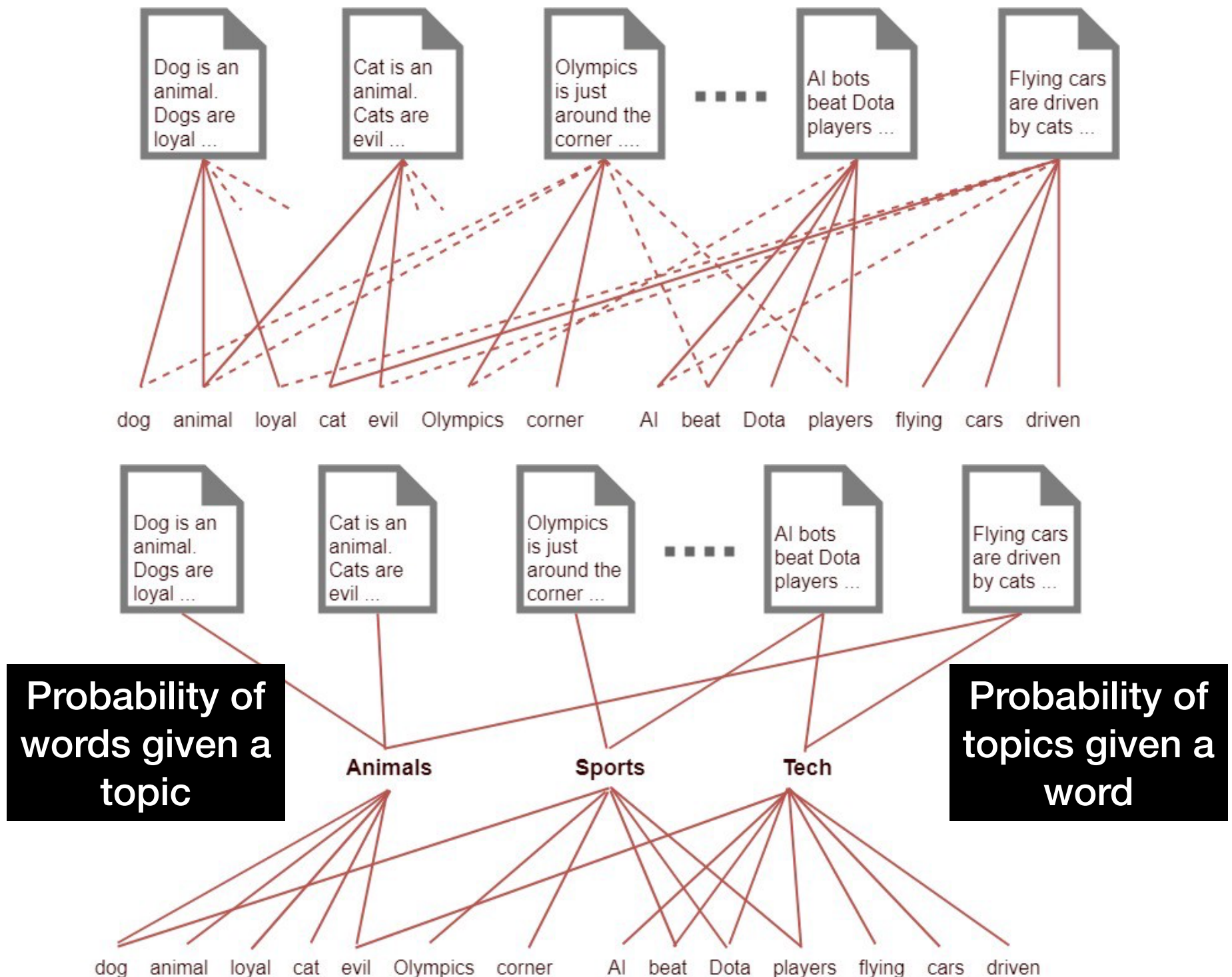
### Top words:

fossiele\_brandstof,  
fossiele\_brandstoffen,  
brandstoffen, energie, opwarming,  
opwarming\_van\_de\_aarde, aarde,  
duurzame\_energie,  
energiebronnen, brandstof



# Topic modeling

1. Input is a collection of documents, in which particular words occur (e.g., 'dog', 'animal').
  2. Create a vocabulary of individual words, which are treated as occurring independently in a document.
  3. Map words to any kind of arbitrary intrinsic representation of a topic and then map a document in which a word occurs to this topic on the basis of how many of words in a text are actually assigned to the same topic.
  4. Iteratively increase the relationship between a word and a hidden topic and use this to pool a document to the hidden topic.
- => Contrastive learning on the basis of words and contrastive learning that takes place on the basis of documents.



# Latent Dirichlet Allocation (LDA)

Vayanski and Kumar, 2020

- Generative approach: how likely is a document for a hidden/latent topic
- Matrix factorization technique: nr. of topics **K** need to be set in advance
  - A collection of documents can be represented as a document-word matrix
  - From the document-word matrix, two reduced matrices are derived:  
document-topic and topic-word

document-word/term  
matrix

|    | W1 | W2 | W3 | Wn |
|----|----|----|----|----|
| D1 | 0  | 2  | 1  | 3  |
| D2 | 1  | 4  | 0  | 0  |
| D3 | 0  | 2  | 3  | 1  |
| Dn | 1  | 1  | 3  | 0  |

*Nr. of topics  
k is fixed*

|    | K1 | K2 | K3 | K |
|----|----|----|----|---|
| D1 | 1  | 0  | 0  | 1 |
| D2 | 1  | 1  | 0  | 0 |
| D3 | 1  | 0  | 0  | 1 |
| Dn | 1  | 0  | 1  | 0 |

document-  
topic  
matrix

|    | W1 | W2 | W3 | Wm |
|----|----|----|----|----|
| K1 | 0  | 1  | 1  | 1  |
| K2 | 1  | 1  | 1  | 0  |
| K3 | 1  | 0  | 0  | 1  |
| K  | 1  | 1  | 0  | 0  |

topic-  
word/term  
matrix

- Random initialisation of the reduced matrices is improved using sampling:
  - the proportion of words in a document **D** that is assigned to topic **K**
  - the proportion a word **W** is assigned topic **K** over all documents in which **W** occurs
- How well for a particular topic the probability of a word can be predicted. By optimizing the probability and iterating over the data, it clusters the documents on the basis of the word associations over the topics.



# Limitations of LDA and alternatives

---

- Fixed topics —> run many times to find optimal number of topics
- Random initialisation and internal sampling —> results differ for each run
- Topics independent from each other —> larger texts exhibit mixed topics (e.g., ‘finance and sports’ not taken into account)
- Words are treated as if independent of each other —> bag-of-words but semantic drift across documents (queen’s gambit, queen’s throne)
- Correlated Topic Model (CTM) —> words from related topics more probable to co-occur within topics (co-variance matrix): to solve topics independence problem (Vayanski and Kumar, 2020)
- Hierarchical LDA —> directed graph of topics (grouping to high-level topics to solve final set of topics) (Vayanski and Kumar, 2020)

None of these approaches works out-of-the-box

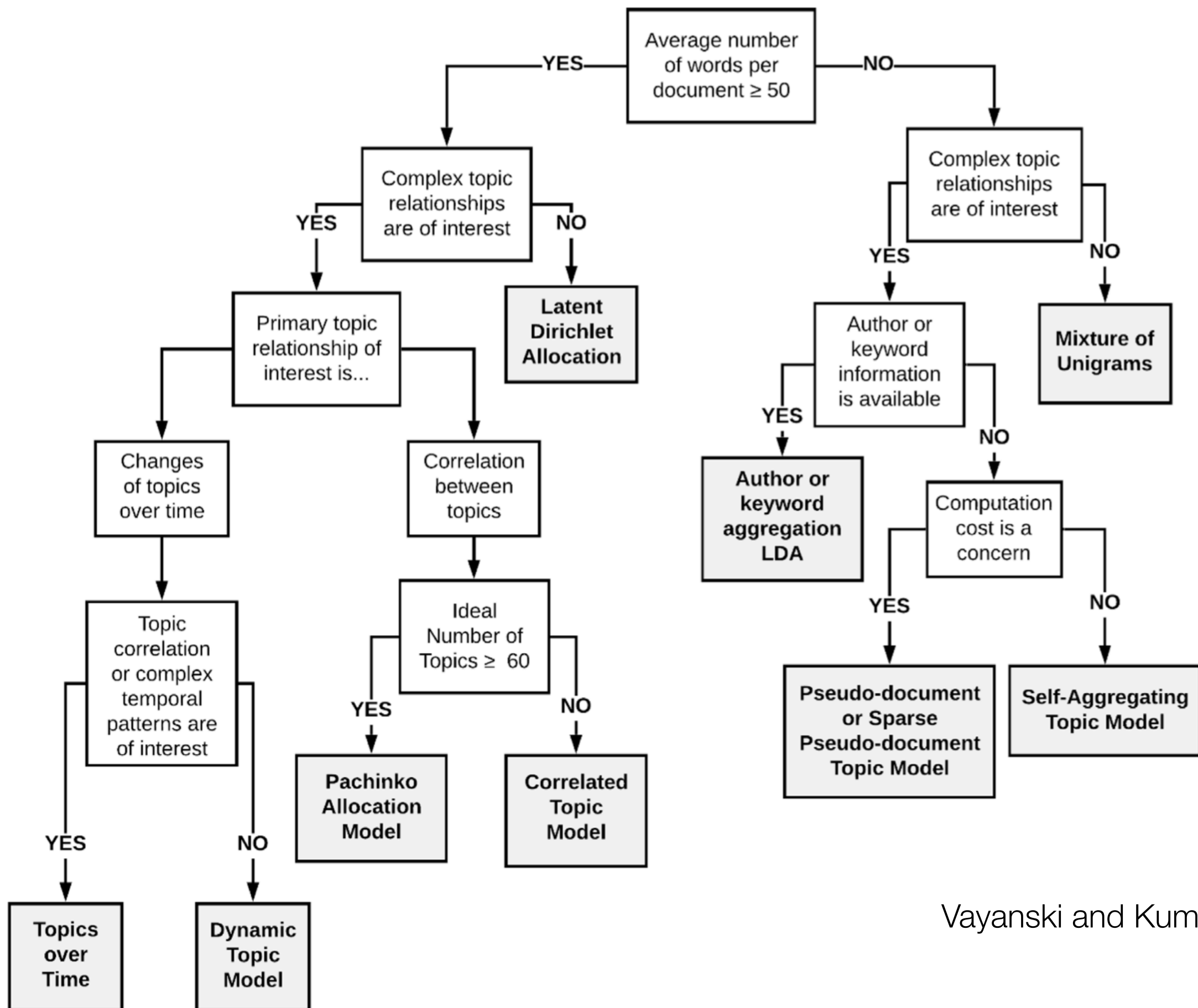
# Limitations of LDA and alternatives

---

- Embedding based representations: for every word add related words to topics (solves data sparseness)
- Short-text topic models:
  - each message has a single topic (few words associated with a single topic);
  - use of meta data such as hashtags, geocodes, user names, dates;
  - noisy and almost no co-occurrence information (multiple words voting for the same topic)
- Time-based topic modeling —> temporal buckets or continuous rise and fall
- Dynamic topic modeling -> if the topics change over time (temporal patterns are not of interest)

| Topic Model Comparison |             |             |              |           |              |        |          |               |      |
|------------------------|-------------|-------------|--------------|-----------|--------------|--------|----------|---------------|------|
|                        | Methodology |             |              |           | Designed For |        |          | Document Type |      |
| Model                  | Generative  | Graph-based | Matrix-based | NLP-aided | Static       | Online | Temporal | Short         | Long |
| LSI [23]               |             |             | x            |           | x            |        |          |               | x    |
| pLSI [30]              | x           |             |              |           | x            |        |          |               | x    |
| DMM [57]               | x           |             |              |           | x            |        |          |               | x    |
| LDA [8]                | x           |             |              |           | x            |        |          |               | x    |
| HDP [74]               | x           |             |              |           | x            |        |          |               | x    |
| CTM [40]               | x           |             |              |           | x            |        |          |               | x    |
| DTM [7]                | x           |             |              |           |              |        | x        |               | x    |
| TOT [86]               | x           |             |              |           |              |        | x        |               | x    |
| SWB [15]               | x           |             |              |           | x            |        |          |               | x    |
| Online LDA [4]         | x           |             |              |           | x            | x      |          |               | x    |
| Online LDA [29]        | x           |             |              |           | x            | x      |          |               | x    |
| Online HDP [83]        | x           |             |              |           | x            | x      |          |               | x    |
| MTTM [52]              | x           |             |              |           |              |        | x        |               | x    |
| cDTM [82]              | x           |             |              |           |              |        | x        |               | x    |
| NMF [69]               |             |             | x            |           | x            |        |          |               | x    |
| NMF [36]               |             |             | x            |           |              |        | x        | x             | x    |
| NMF [89]               |             |             | x            |           | x            |        |          | x             |      |
| ETM [13]               |             | x           |              |           |              |        | x        | x             |      |
| TS [22]                |             | x           |              |           | x            |        |          | x             |      |
| TFM [21]               |             | x           |              |           |              | x      | x        | x             |      |
| PTM [18]               |             | x           |              |           | x            |        |          | x             |      |
| BTM [88]               | x           |             |              | x         | x            |        |          | x             | x    |
| SATM [62]              | x           |             |              | x         | x            |        |          | x             |      |
| LF-DMM [56]            | x           |             |              | x         | x            |        |          | x             |      |
| LF-LDA [56]            | x           |             |              | x         | x            |        |          | x             |      |
| NVDM [49]              | x           |             |              | x         | x            |        |          |               | x    |
| lda2vec [51]           | x           |             |              | x         | x            |        |          | x             |      |
| ETM [60]               | x           |             |              | x         | x            |        |          | x             |      |
| GSDMM [94]             | x           |             |              |           | x            |        |          |               | x    |
| GPUDMM [42]            | x           |             |              | x         | x            |        |          | x             |      |
| GPUPDMM [42]           | x           |             |              | x         | x            |        |          | x             |      |
| DREx [6]               | x           |             |              | x         | x            |        |          |               | x    |
| CSTM [44]              | x           |             |              | x         | x            |        |          |               | x    |
| WELDA [12]             | x           |             |              | x         | x            |        |          |               | x    |
| LapDMM [45]            | x           |             |              | x         | x            |        |          | x             |      |
| CluWords [78]          |             |             | x            | x         | x            |        |          | x             |      |
| CluHTM [79]            |             |             | x            | x         | x            |        |          | x             |      |
| ETM [25]               | x           |             |              | x         | x            |        |          |               | x    |
| D-ETM [26]             | x           |             |              | x         |              |        | x        |               | x    |
| TND [20]               | x           |             |              | x         | x            |        |          | x             |      |

Table 1. Topic Model Design Characteristics by Model. Ordered by appearance in this survey.



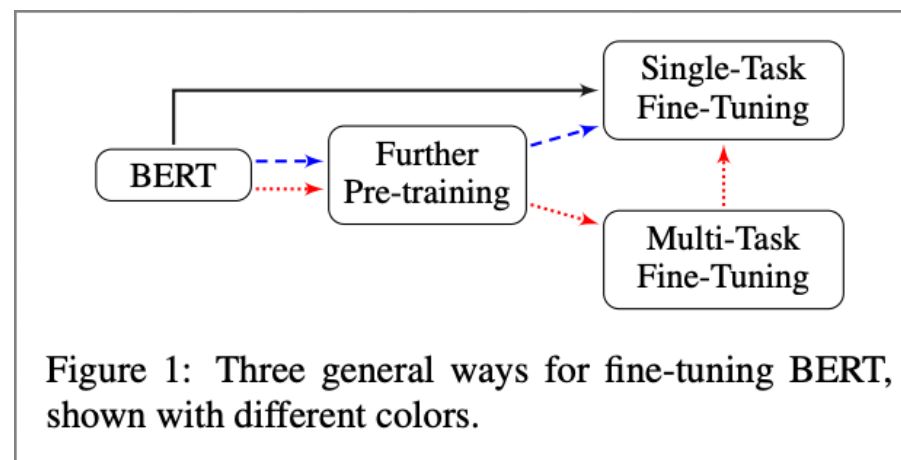
Vayanski and Kumar, 2020

**Fig. 8.** Decision tree for selecting a topic modeling method.

# Topic classification using pretrained models

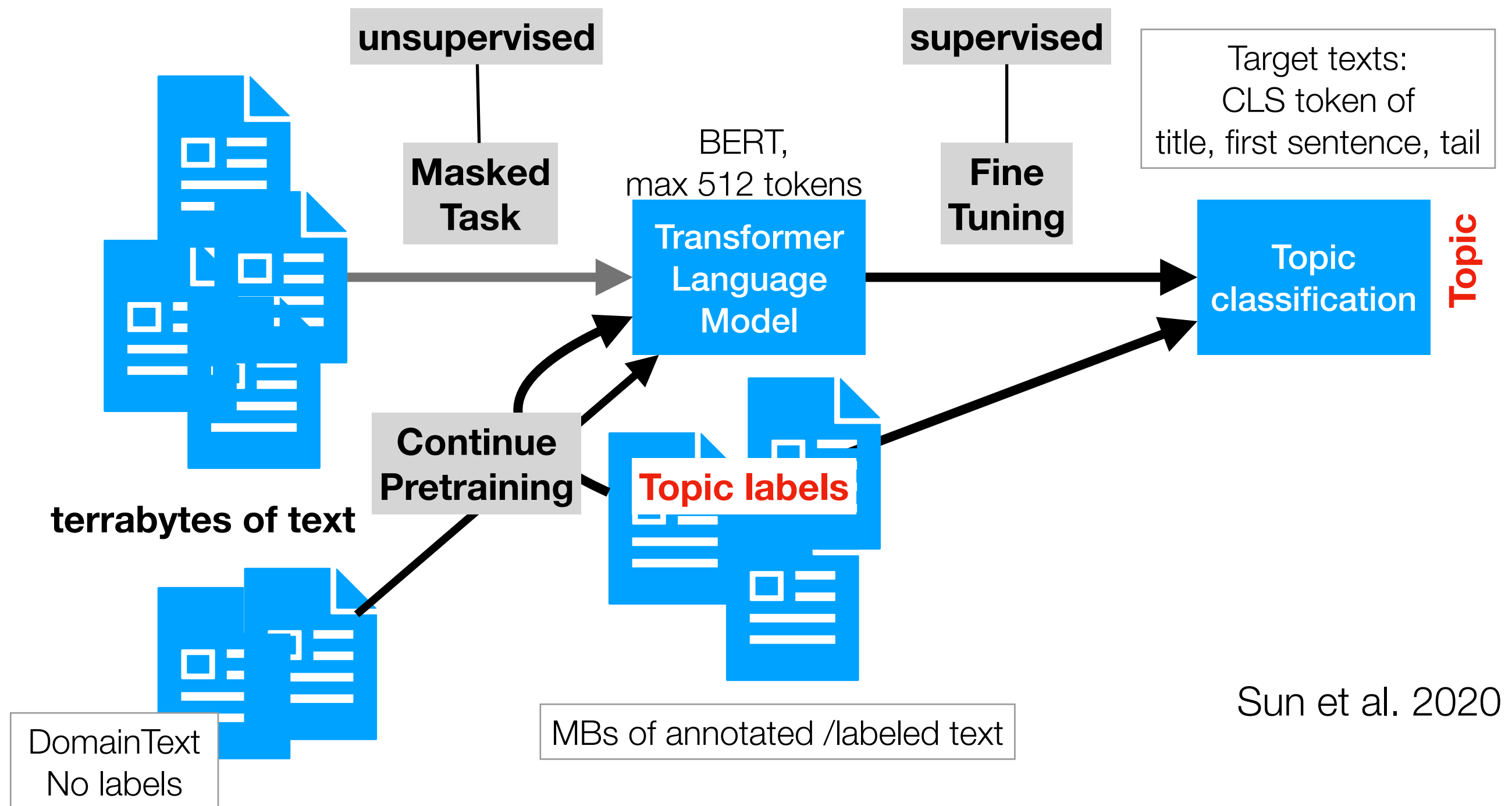
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- Various examples in *Sun et al. 2020, arxiv, How to Fine-Tune BERT for Text Classification?*
  - <https://github.com/xuyige/BERT4doc-Classification>

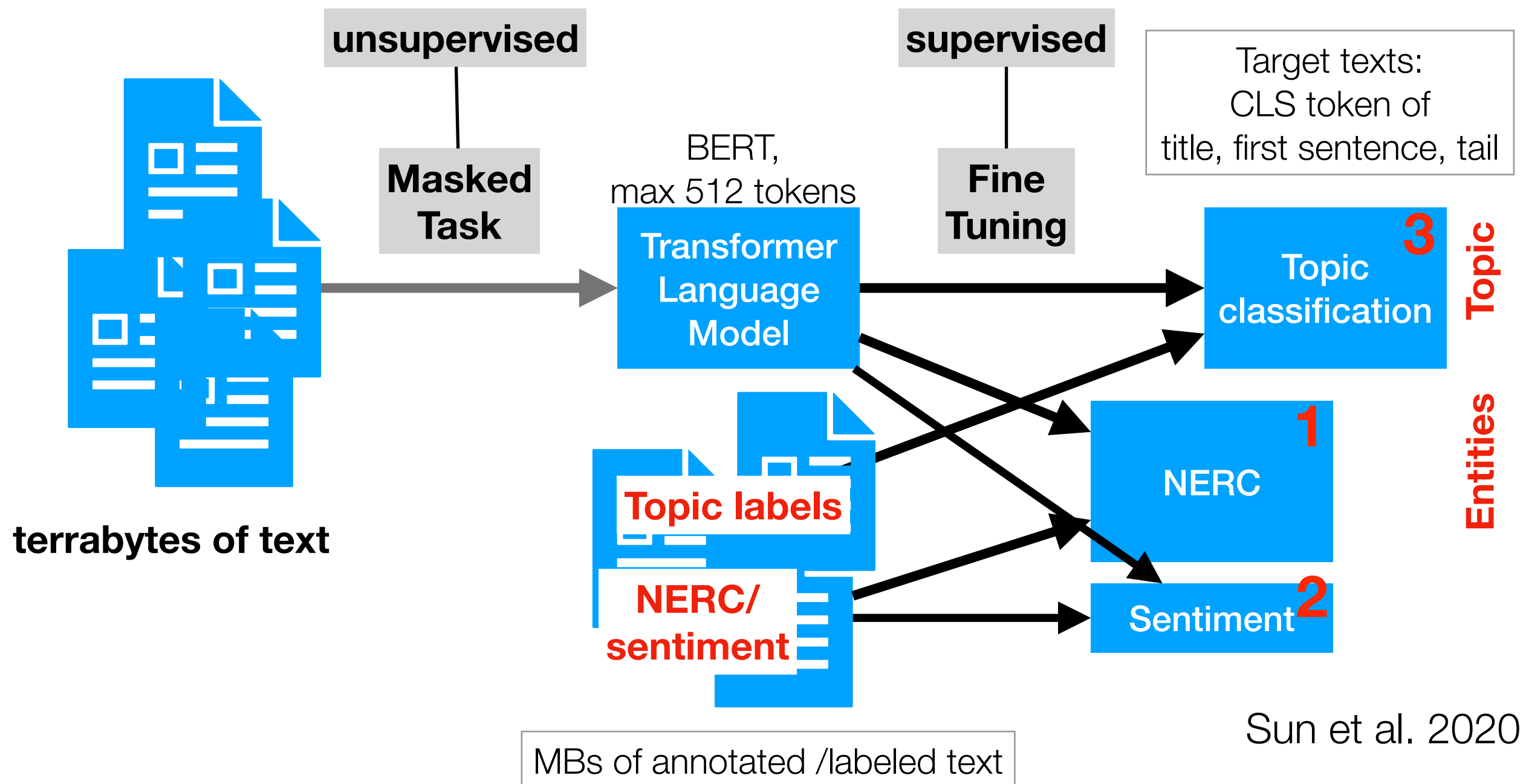


- (1) further pre-train BERT on within-task training data or in-domain data;
- (2) optional fine-tuning BERT with multi-task learning if several related tasks are available; by training for other tasks the internal representations in a model will be adapted to capture those aspects of a text as well
- (3) fine-tune BERT for the target task.

# Fine tuning & continued pretraining



# Fine tuning on multiple tasks





# Fine tuning on multiple data sets & tasks

| Dataset        | Classes | Type      | Average lengths | Max lengths | Exceeding ratio | Train samples | Test samples |
|----------------|---------|-----------|-----------------|-------------|-----------------|---------------|--------------|
| IMDb           | 2       | Sentiment | 292             | 3,045       | 12.69%          | 25,000        | 25,000       |
| Yelp P.        | 2       | Sentiment | 177             | 2,066       | 4.60%           | 560,000       | 38,000       |
| Yelp F.        | 5       | Sentiment | 179             | 2,342       | 4.60%           | 650,000       | 50,000       |
| TREC           | 6       | Question  | 11              | 39          | 0.00%           | 5,452         | 500          |
| Yahoo! Answers | 10      | Question  | 131             | 4,018       | 2.65%           | 1,400,000     | 60,000       |
| AG's News      | 4       | Topic     | 44              | 221         | 0.00%           | 120,000       | 7,600        |
| DBPedia        | 14      | Topic     | 67              | 3,841       | 0.00%           | 560,000       | 70,000       |
| Sogou News     | 6       | Topic     | 737             | 47,988      | 46.23%          | 54,000        | 6,000        |

Table 1: Statistics of eight text classification datasets. The exceeding ratio means the percentage of the number of samples with a length exceeding 512.

Sentiment analysis, question classification, and topic classification data

# Text fragments and layers

| Layer                   | Test error rates(%) |
|-------------------------|---------------------|
| Layer-0                 | 11.07               |
| Layer-1                 | 9.81                |
| Layer-2                 | 9.29                |
| Layer-3                 | 8.66                |
| Layer-4                 | 7.83                |
| Layer-5                 | 6.83                |
| Layer-6                 | 6.83                |
| Layer-7                 | 6.41                |
| Layer-8                 | 6.04                |
| Layer-9                 | 5.70                |
| Layer-10                | 5.46                |
| Layer-11                | <b>5.42</b>         |
| First 4 Layers + concat | 8.69                |
| First 4 Layers + mean   | 9.09                |
| First 4 Layers + max    | 8.76                |
| Last 4 Layers + concat  | 5.43                |
| Last 4 Layers + mean    | 5.44                |
| Last 4 Layers + max     | <b>5.42</b>         |
| All 12 Layers + concat  | 5.44                |

Table 3: Fine-tuning BERT with different layers on IMDB dataset.

| Method               | IMDb        | Sogou       |
|----------------------|-------------|-------------|
| head-only            | 5.63        | 2.58        |
| tail-only            | 5.44        | 3.17        |
| head+tail            | <b>5.42</b> | <b>2.43</b> |
| hier. mean           | 5.89        | 2.83        |
| hier. max            | 5.71        | 2.47        |
| hier. self-attention | 5.49        | 2.65        |

Table 2: Test error rates (%) on IMDb and Chinese Sogou News datasets.

Last (top) layer is the most effective layer for the text classification task

# Pre-training in-domain & cross-domain

| Domain        | sentiment   |             |              | question    |              | topic       |             |
|---------------|-------------|-------------|--------------|-------------|--------------|-------------|-------------|
| Dataset       | IMDb        | Yelp P.     | Yelp F.      | TREC        | Yah. A.      | AG's News   | DBPedia     |
| IMDb          | <b>4.37</b> | 2.18        | 29.60        | 2.60        | 22.39        | 5.24        | 0.68        |
| Yelp P.       | 5.24        | 1.92        | 29.37        | 2.00        | 22.38        | 5.14        | <b>0.65</b> |
| Yelp F.       | 5.18        | 1.94        | 29.42        | 2.40        | 22.33        | 5.43        | <b>0.65</b> |
| all sentiment | 4.88        | <b>1.87</b> | 29.25        | 3.00        | 22.35        | 5.34        | 0.67        |
| TREC          | 5.65        | 2.09        | 29.35        | 3.20        | 22.17        | 5.12        | 0.66        |
| Yah. A.       | 5.52        | 2.08        | 29.31        | <b>1.80</b> | 22.38        | 5.16        | 0.67        |
| all question  | 5.68        | 2.14        | 29.52        | 2.20        | <b>21.86</b> | 5.21        | 0.68        |
| AG's News     | 5.97        | 2.15        | 29.38        | 2.00        | 22.32        | <b>4.80</b> | 0.68        |
| DBPedia       | 5.80        | 2.13        | 29.47        | 2.60        | 22.30        | 5.13        | 0.68        |
| all topic     | 5.85        | 2.20        | 29.68        | 2.60        | 22.28        | 4.88        | <b>0.65</b> |
| all           | 5.18        | 1.97        | <b>29.20</b> | 2.80        | 21.94        | 5.08        | 0.67        |
| w/o pretrain  | 5.40        | 2.28        | 30.06        | 2.80        | 22.42        | 5.25        | 0.71        |

Table 5: Performance of in-domain and cross-domain further pre-training on seven datasets. Each was further pre-trained for 100k steps. The first column indicates the different further pre-training dataset. “all sentiment” means the dataset consists of all the training datasets in sentiment domain. “all” means the dataset consists of all the seven training datasets. Note that some of the data in Yelp P. and Yelp F. are overlapping, e.g., part of the data in the test set of Yelp P. will appear in the training set of Yelp F., so we remove this part of data from the training sets during further pre-training.

# Multitask & % of data

| Method             | IMDb        | Yelp P.     | AG          | DBP         |
|--------------------|-------------|-------------|-------------|-------------|
| BERT-FiT           | 5.40        | 2.28        | 5.25        | 0.71        |
| BERT-MFiT-FiT      | 5.36        | 2.19        | 5.20        | 0.68        |
| BERT-CDPT-FiT      | 5.18        | <b>1.97</b> | <b>5.08</b> | <b>0.67</b> |
| BERT-CDPT-MFiT-FiT | <b>4.96</b> | 2.06        | 5.13        | <b>0.67</b> |

Table 7: Test error rates (%) with multi-task fine-tuning.

ITPT = within-task pretraining

FiT = fine-tuning

MFiT = multitask

CDPT = cross-domain pretraining

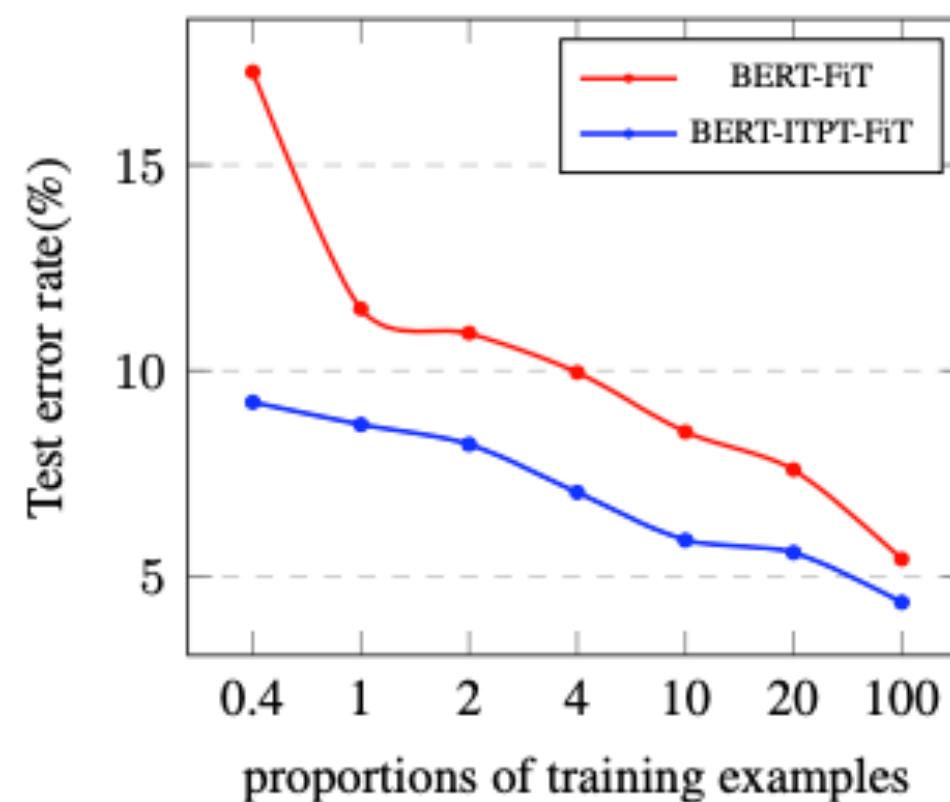


Figure 4: Test error rates(%) on IMDb dataset with different proportions of training examples.

# Lessons learned fine-tuning BERT

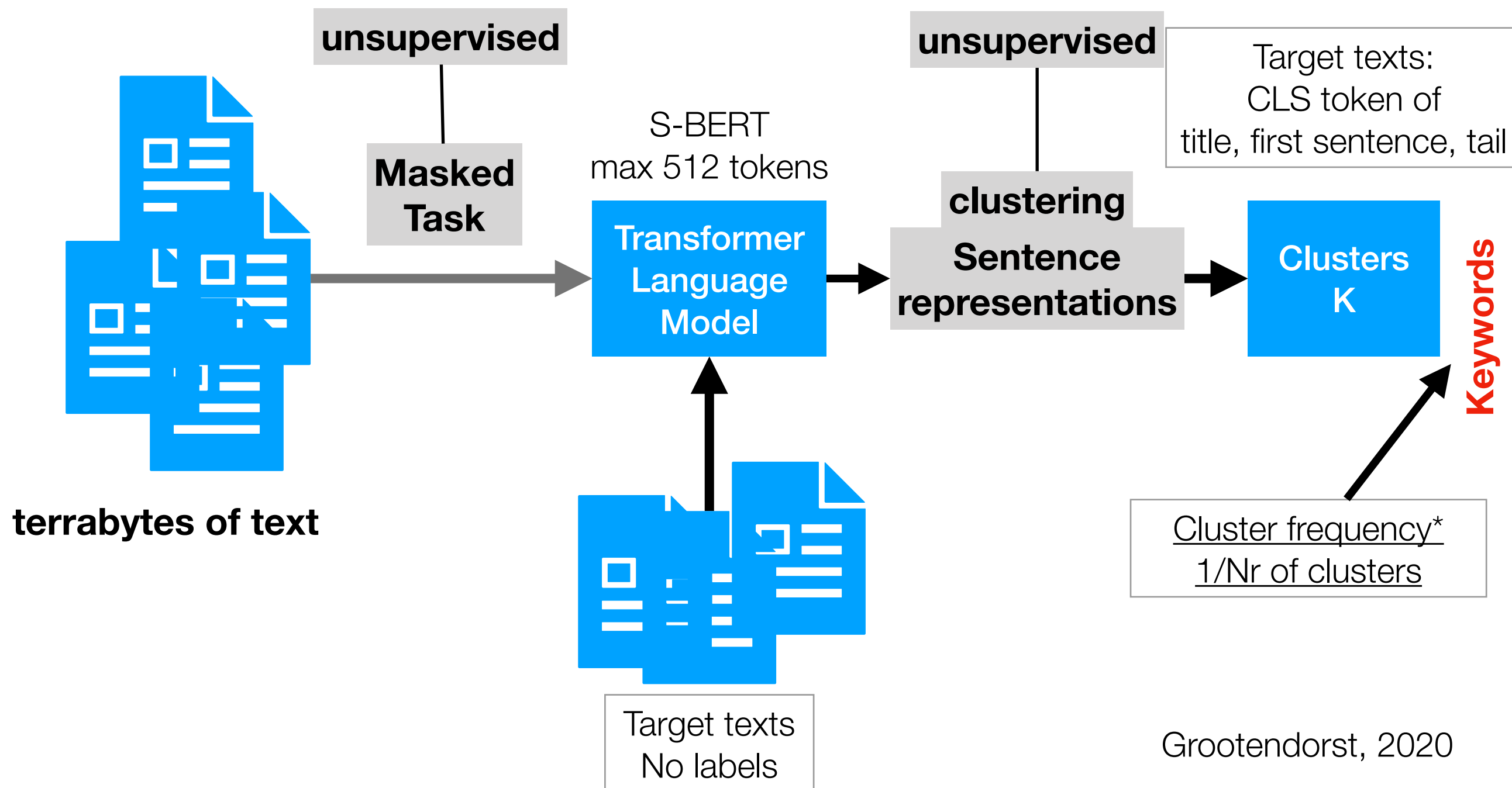
- Top layer(s) most useful for text classification (top layers capture semantic info and lower layers morphological or syntactic info)
- Decreasing learning-rate to overcome catastrophic forgetting (when pretrained weights get ignored by fine-tuning): if the model takes big steps, it can completely change the internal representations learned in pretraining
- Within-task and in-domain pretraining boosts performance
- Multi-task fine-tuning helps marginally
- Fine-tuning already effective with little training data

# Topic modelling using pretrained models

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- Another example:
  - Grootendorst, Maarten. "Bertopic: Leveraging bert and c-tf-idf to create easily interpretable topics." URL <https://doi.org/10.5281/zenodo.4381785> (2020).
  - <https://github.com/MaartenGr/BERTopic>
- (1) represent a text as 512 tokens using a contextual transformer model (S-BERT has fine-tuned CLS token to represent a sentence)
- (2) apply clustering (e.g. LDA) to learn a relation between embedding representations and topics
- (3) extract keywords from cluster using TF/ClusterFrequency

# Clustering representations from pretrained models





# Evaluation

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- Explicitly labelled documents (ground-truth topics)
- Eye balling (qualitative):
  - Keyword extraction and inspection
  - Visualisation (National Research Agenda, Newsbrief): graph with connections
  - Reading samples per cluster
- Intrinsic measurement (quantitative) of clusters' properties



# Evaluation

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- Topic recall: are all topics in a collection identified to the true degree (ground truth); are all topics in a collection are identified?
- Accuracy: proportion of documents correctly labeled for a topic (ground truth)
- Purity: accuracy considering the dominant (highest scoring) topic
- KL-divergence: difference in topic probability distribution model A against model B (B could be ground through or another approach)
- Perplexity: how well can a model predict hold-out sample documents given a topic (generative models)

# Evaluation

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- Coherence measures: how noisy or unintelligible are topics
  - precision: proportion of top-scoring words (keywords) from approximated topic in ground-truth topic
  - pointwise mutual information (PMI): co-occurrence frequency of words in a topic higher than expected given their individual overall frequency (no ground-truth needed); measures the frequency of the combination of two words relative to the frequency of the individual words.
  - diversity: word overlap across topics, percentage of unique words in the topic set given the top-k words of each topic (no ground-truth needed); how many words are unique for a topic and don't occur in other topics

# Factors to consider when choosing a topic model

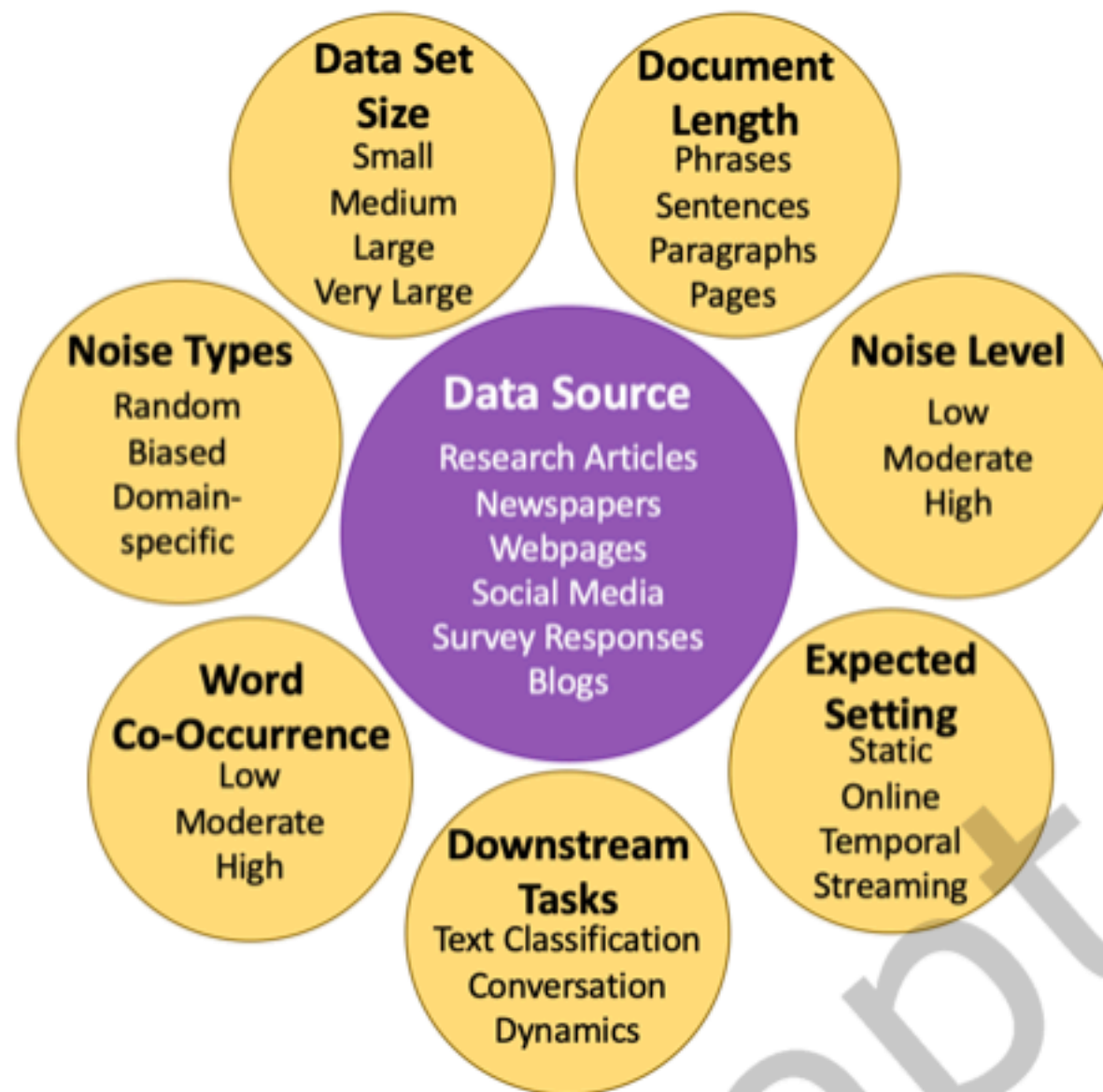


Fig. 6. Factors to consider when choosing a topic model

# Master's programs

# Faculty of Humanities

- **Language and AI** (1 year)
- **Research Master's in Linguistics** (Human Language Technology track) (2 years)